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Pulse Recovery Procedure

This document outlines the available options to recover Pulse following a catastrophic event or outage that disrupts all Pulse nodes.

Recovery focuses on restoring the system to a state where Pulse can process and score real-time payments and digital events, ensuring both real-time and scheduled metrics are accurate so that the DS/rule/model performance is as expected.



Please note, the timings in this guide are relative to the time it takes a Pulse node to publish. The figures used were accurate as of April 2024, but may vary over time.

However, while the exact timings may differ, the principles remain the same.

Pulse operates with a three-node setup, where a single node requires approximately **1h 45m** to complete a publish. The following recovery strategies detail how to handle such scenarios effectively.

Recovery Strategies

Option 1: Abort Publish and Perform Manual Publish

Process:

1. Access the Pulse UI and abort the publish manually.
2. Ensure all three nodes are part of the cluster.
3. Initiate a parallel (non-rolling) publish for all nodes simultaneously.

Recovery Timeline:

- **First 1h 50m:** No processing or scoring occurs while the abort and re-publish steps are performed. The timeline includes an additional ~5 minutes required to abort and initiate the manual re-publish.
- **After 1h 50m:** Full service and redundancy are restored simultaneously.



Note that parallel publishing across all three nodes may take slightly longer if database performance becomes a bottleneck. Pre-production testing is recommended to validate this behavior.

Considerations:

- Events processed by the GCP load-balancer during the outage will likely have incorrect scores due to:
 - **Real-time metrics** being absent or incorrect.
 - **Workflow clock misalignment**, which could negatively affect model scores and alert rates.
- Despite these issues, previous testing has shown that the impact on scores and alert rates is manageable.

Manually publish with recovery skipped, then republish

Process:

1. Access the Pulse UI and abort the publish manually.
2. Ensure all 3 Pulse server nodes are part of the cluster, and perform a parallel (non-rolling) publish with "skip recovery" selected.

3. After the publish, save the workflow without changes.
4. Perform a rolling publish (with application restart selected, without **skip recovery**)

Recovery Timeline:

- **First 5-10 minutes:** Pulse will not be processing or scoring the events.
- **After 1h 45m:**
 - Partial service and full redundancy are restored at the same time.
 - Pulse will score events, but it will consider all real-time metrics as having the value zero.
 - All scheduled metrics will be correctly updated.
 - Model scores/performance and alert rates will be impacted.
- **After another 1h 45m (total 1h 55m from the start):**
- Half the transactions will have totally correct scores, and the other half will continue to have incorrect real-time metrics.



This is because after the first node publishes, one of the remaining two will go down. This means there are two nodes scoring at this point - one that just finished publishing, fully correct, and one with correct SMJs but incorrect real-time metrics.

- **After another 1h 45m (total ~3h 40m from the start):**
- Full service recovery is complete with correct real-time and scheduled metrics.



Full redundancy has not yet been achieved as 2 of 3 nodes will be scoring, while the 3rd will go down to update itself.

- **After another 1h 45m (total ~5h 25m from the start):** Full redundancy is achieved.



All the nodes are up at this and correctly updated.

Considerations:

- This option ensures quicker partial service restoration compared to Option 1, with limited impact on metrics during the recovery process.
- Transactions processed during the initial phase may have incomplete real-time metrics, but overall service availability is restored sooner.
- Data Science (DS) team changes cannot be published during the recovery process.

Recommendations

Feedzai recommends Option 2. It is the standard approach followed in the Feedzai cloud. It tends to have the smallest negative impact overall

- The service is mostly restored (with just real-time metrics missing) very quickly
- Followed by full restoration in steps (half the nodes fully recovered after another 1h 45m, and full restoration after another 1h 45m)